Producer–Consumer Communication Using Anonymous Pipes

Operating Systems IPC Experiment

This document contains the submitted Linux shell program and execution screenshots. The shown program implements a simple command-line shell using C, while the execution demonstrates commands entered and the program running successfully in Ubuntu/Linux.

Source Code Screenshot

```c
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        // Display prompt
        printf("myshell> ");
        fflush(stdout);

        // Read user input
        if (fgets(input, sizeof(input), stdin) == NULL) {
            printf("\nExiting...\n");
            break;
        }

        // Remove newline character
        input[strcspn(input, "\n")] = '\0';

        // Handle exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }
}
}
```

Program Execution Screenshot

```
lenovo@SAISHASHANK:~$ gcc skill2.c -o skill2
lenovo@SAISHASHANK:~$ ./skill2
myshell> ls
myshell> pwd
myshell> date
myshell> whoami
myshell> ^C
lenovo@SAISHASHANK:~$ nano skill2.c
lenovo@SAISHASHANK:~$ 
```

Result

The C program was compiled using GCC and executed successfully. The shell accepted commands such as **ls**, **pwd**, **date**, and **whoami**. The screenshots are included as evidence of compilation and execution.